

Digital Parking Permit System – Final Practical Report

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Course Name: SER216 – Software Engineering

Project Name: Digital Parking Permit System

Introduction: This project presents a Digital Parking Permit System designed for a university environment. The system aims to simplify and digitize the process of vehicle registration, parking permit requests, and parking verification. It replaces manual processes with an automated workflow that improves efficiency, accuracy, and security for both users and administration.

System Requirements: Functional Requirements The system allows students and staff to register their vehicles by providing necessary details such as vehicle number and type. It enables users to request parking permits and view available parking zones. Additionally, the system supports permit approval processes and allows security personnel to verify permits and record violations.

Non-Functional Requirements The system must respond to user actions within two seconds to ensure efficiency. It must also provide secure access through authentication mechanisms to protect user data and system integrity.

Workflow Overview: The system follows a structured workflow starting from vehicle registration, followed by permit request submission. The system then checks available parking zones and processes permit approval based on validity. Finally, security staff verify permits during parking checks. This workflow ensures a smooth and logical process from start to finish.

Testing Strategy: The system was tested using system testing, acceptance testing, and regression testing. System testing ensured that all components function together correctly. Acceptance testing verified that the system meets user expectations. Regression testing ensured that new updates do not break existing functionality.

Defect Analysis: A defect was identified where a valid parking permit was incorrectly displayed as expired during verification. This issue is categorized as a logic error with high severity and priority. The possible cause is incorrect date validation or system time mismatch. A regression test was designed to validate permit status after fixing the issue.

Tools Used: The project utilized Google Docs for documentation, Google Sheets for test case management, diagrams.net for workflow diagram creation, and GitHub for version control and project tracking.

Conclusion: The Digital Parking Permit System demonstrates how software engineering principles can be applied to solve real-world problems efficiently. The system improves accuracy, reduces manual effort, and enhances user experience through a structured and test-driven approach.